

HASSAN SAEED HASSAN ALBATTRA

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Summary

DevOps Engineer with MSc-level training in cloud computing and AI-native software engineering, focused on cloud infrastructure, CI/CD automation, containerized deployments, and security-first pipeline design. Hands-on experience provisioning AWS infrastructure with Terraform, deploying containerized ML models on EC2, and building end-to-end cloud pipelines (S3, EMR, VPC, IAM). Completing 180-hour AWS DevOps professional training toward AWS Developer and DevOps Engineer Associate certifications.

Education

Queen's University, Kingston, ON, Canada Jan. 2026 – Present

MSc – Computer Software Engineering | Digilians Initiative — Fully funded national scholarship by Egypt's MCIT

Focus Areas: DevOps, MLOps, CI/CD, GitOps, Kubernetes, Infrastructure as Code, Cloud Computing, Generative AI, LLM applications.

Courses: CISC 814 (DevOps & MLOps), CISC 886 (Cloud Computing), CSAI 810 (GenAI), CSAI 801 (AI/ML), CISC 818 (AI-Native SE)

Fayoum University, Fayoum, Egypt Sep. 2023 – Dec. 2025

MSc – Electrical Engineering (Computer & Systems Engineering)

Thesis: Phishing URL detection using ML/DL — built a Dockerized, ONNX-optimized inference pipeline meeting real-time SLAs. Published open-source code (GitHub) and dataset (Zenodo, 339K URLs).

Misr University for Science and Technology (MUST), Giza, Egypt Sep. 2019 – Feb. 2023

BEng – Electrical Engineering (Computer & Software Engineering) · GPA: 3.4/4.0 (85.2%) · **Ranked 1st in Class**

Experience

Misr University for Science and Technology (MUST) Feb. 2023 – Present

Teaching Assistant – Computer & Software Engineering Dept.

Giza, Egypt

- Delivered lectures and hands-on labs for 300+ students/semester across Data Security (CSE566), Cryptography (CSE565), and Computer Network Security (CSE567).
- Designed 15+ hands-on labs on encryption algorithms, secure coding, ethical hacking, and threat modeling; supervised 10+ capstone projects on secure system deployments.
- Mentored students on Git workflows, automated testing, and secure SDLC practices; 3 students presented security research at departmental symposia.

Projects

DevOps Sentiment K8s GitOps – CI/CD & GitOps Pipeline (CISC 814) 2026

- Built a complete **Gitea Actions CI pipeline** for a FastAPI/TextBlob ML microservice: automated linting (**ruff**), static type checking (**mypy**), pytest with coverage, **multi-stage Docker build** (non-root, python:3.11-slim), image push to private registry, and **SBOM generation** (CycloneDX).
- Deployed to **k3d Kubernetes** via **ArgoCD GitOps** with canary release strategy; configured **Prometheus + Grafana** observability dashboard and OpenTelemetry instrumentation. [\[GitHub\]](#)

Cyber-SOC-Chatbot – End-to-End Cloud LLM on AWS (CISC 886) 2026

- Provisioned full AWS infrastructure with **Terraform** (VPC, public/private subnets, IGW, security groups, IAM roles, S3, EMR) following the Well-Architected Framework.
- Preprocessed 20M+ cybersecurity events using **PySpark on AWS EMR**; fine-tuned TinyLlama-1.1B with **QLoRA/PEFT**, exported to GGUF, deployed via **Ollama on EC2 g4dn.xlarge** with systemd.
- Served model through **OpenWebUI (Docker)** with browser chat; configured security groups for SSH, API, and UI. [\[GitHub\]](#)

EdgePhish-5G – Dockerized ML Security Pipeline 2024 – 2025

- Containerized (Docker) a hybrid BERT-TF-IDF phishing detection pipeline; ONNX Runtime optimization reduced P99 latency from 17.2 ms to 7.3 ms (2.36×), meeting ≤10 ms SLA.
- Implemented reproducible, single-command deployment with environment-variable config; published on [\[GitHub\]](#) and [\[Zenodo\]](#) (339K URLs).

GuardedRAG – Safe Retrieval-Augmented Generation System (CSAI 810) 2026

- Built a secure RAG pipeline: PDF ingestion, LangChain chunking, Sentence Transformer embeddings, and semantic retrieval from **ChromaDB**.
- Orchestrated **Gemini 2.5 Flash + OpenAI Moderation API** in a 5-stage safety pipeline (scope check → moderation → retrieval → generation → output check); deployed via Gradio. [\[GitHub\]](#) · [\[Live Demo\]](#)

Crime Type Classification – ML for Police Resource Allocation (CSAI 801) 2026

- Built a leakage-aware, chronological ML pipeline (241K Toronto crime records, 393 features) comparing 8 model configurations; XGBoost with inverse-frequency class weighting achieved best macro F1 (0.426) across 5 crime types under severe class imbalance.
- Engineered cyclical temporal features, one-hot spatial encoding, and evaluated SMOTE resampling and MLP baselines

with GridSearchCV + TimeSeriesSplit. [\[GitHub\]](#)

Technical Skills and Training

Languages	Python, Bash/Shell, Java, C/C++, JavaScript, SQL, YAML/JSON
Cloud / DevOps	AWS, Docker, Kubernetes, Terraform, CloudFormation, Ansible, CI/CD, GitHub Actions, Jenkins, ArgoCD, GitOps, Linux, IaC
AWS Services	EC2, S3, EMR, Lambda, API Gateway, DynamoDB, RDS, VPC, IAM, ECS, EKS, SNS, SQS, CloudWatch,
DevSecOps	CloudTrail, Cognito, CodePipeline, CodeBuild, Systems Manager, Elastic Beanstalk, X-Ray Secure pipeline design, secrets management, IAM least privilege, encryption, MFA, threat modeling
AI / ML Deployment	PyTorch, ONNX Runtime, scikit-learn, XGBoost, HuggingFace, QLoRA/PEFT, Ollama, GGUF, RAG, LangChain, Gradio
Monitoring / MLOps	Prometheus, Grafana, MLflow, OpenTelemetry, vLLM, DVC, model versioning
Databases / Tools	PostgreSQL, MongoDB, ChromaDB, Git, GitHub, Jupyter, VS Code
Certifications (in progress): AWS Developer Associate · AWS DevOps Engineer Associate	
Training: AWS DevOps Track (180 hrs, NTI) — Cloud Practitioner · Cloud Developer · Cloud DevOps Offensive Cybersecurity (B.S.A) · Cybersecurity II (Security Meter) · Embedded Systems 80 hrs (AMIT) · Web Development 60 hrs (New Horizons)	

Publication

H. AlBattrra, R. Abul Seoud, D. A. Salem, “EdgePhish-5G: Hybrid BERT–TF-IDF Phishing URL Detection for Real-Time 5G Edge Deployment,” *Submitted to ITC*, 2026. [\[Dataset DOI\]](#)